

Lecture 09 SQL WorkSheet

Create three tables Emp , Dept and Location to store details of Employees and Departments with below fields.

Emp (empId , fName , lName ,email ,hireMonth,salary , deptNo)

Dept (deptNo , dName , dLoc , createDate, DepManagerName)

Location (LocID, LocTown, LocProvince)

Employees should be worked in departments which is only stated in Department table. But there can be some employees who are not assigned to any department. Also Department table will have DLocs which is stated under LocID property in location table.

Use below scripts to create and Insert data to above tables

```
CREATE TABLE Location
(
    LocID INT(5),
    LocTown VARCHAR (25),
    LocProvince VARCHAR (25),
    CONSTRAINT l_lno_pk PRIMARY KEY (LocID)
);
```

```
CREATE TABLE Dept
(
    deptNo INT(4),
    dName VARCHAR(20) UNIQUE NOT NULL,
    dLoc INT(5) ,
    createDate TIMESTAMP,
    DepManagerName VARCHAR(25),
    CONSTRAINT d_dno_pk PRIMARY KEY (deptNo),
    CONSTRAINT d_dloc_fk FOREIGN KEY (dLoc) REFERENCES
    Location(LocID)
);
```

```

CREATE TABLE Emp
(
empId INT(6),
fName VARCHAR(50) NOT NULL,
lName VARCHAR(50) NOT NULL,
email VARCHAR(100) UNIQUE NOT NULL,
hireMonth VARCHAR(10),
salary INT,
deptNo INT(4),
CONSTRAINT e_eid_pk PRIMARY KEY (empId),
CONSTRAINT e_dno_fk FOREIGN KEY (deptNo) REFERENCES Dept(deptNo)
);

```

-- Insert Below records to Dept Employee and Location tables

```

INSERT INTO Location (LocID,LocTown,LocProvince)
VALUES
    (100,'Colombo','Western'),
    (200,'Kandy','Central'),
    (300,'Jaffna','Notheren'),
    (400,'Ampara','Eastern'),
    (500, 'Galle','Southern');

```

```

INSERT INTO
Dept (deptNo, dName,dLoc,createDate,DepManagerName)
VALUES
    (10, 'IT', '100','2015-03-08','Mike'),
    (20, 'Operations', '100','2014-03-08','Simon'),
    (30, 'Sales', '400','2015-03-08','Donald'),
    (40, 'Marketing', '200','2016-03-08','Joe'),
    (50, 'Management', '200','2016-03-08','Jim'),
    (60, 'Exams', '300','2017-03-08','Elon'),
    (70, 'Security', '400','2018-05-08','Mike'),
    (80, 'Registry', '300','2018-03-08','Sam');

```

```

INSERT INTO
Emp (empId,fName ,lName,email,hireMonth,salary,deptNo)
VALUES
(1001, 'Jim', 'King', 'jk@firm.com', 'January', 25000, 50),
(1002, 'Jane', 'Queen', 'jq@firm.com', 'February', 35000 , 50),
(1003, 'Jen', 'Probert', 'jp@firm.com', 'January', 50000, 10),
(1004, 'Mike', 'Brent', 'mb@firm.com', 'February', 85000 , 10),
(1005, 'Nadia', 'Tamsa', 'nt@firm.com', 'January', 55000, 20),
(1006, 'Lesley', 'Morgan', 'lm@firm.com', 'February', 90000,
30),
(1007, 'Simon', 'King', 'sk@firm.com', 'January', 75000, 20),
(1008, 'Adam', 'Brent', 'ab@firm.com', 'March', 70000, 30),
(1009, 'Russel', 'Brent', 'rb@firm.com', 'March', 70000, 10),
(1010, 'Paul', 'Brent', 'pb@firm.com', 'March', 70000, 20),
(1011, 'Kyle', 'Ewans', 'ke@firm.com', 'March', 70000, 10);

```

Write down SQL statements for below questions

1. Find out all employees who gets higher salary than Adam Brent;
2. Write a query to display the name (first name and last name), salary, department id for those employees who earn a salary same as the smallest salary of any of the departments.
3. Write a query to display the employee id, employee name (first name and last name) for all employees who earn more than the average salary of the company.
4. Write a query in SQL to display all detailed information of those departments which starting salary is at least 8000.
5. Display all department names where departments minimum salary higher than the minimum salary for department 30
6. Display all the towns which has high employee population than the number of employees who works in Ampara.
7. Display complete department details where employees who earn more than 5000 works in.
8. Display all employee details who joined in same month and work in same department as the employee id 1011